

Changes in ultrasound-assessed carotid intima-media thickness and plaque with a Mediterranean diet: a substudy of the PREDIMED trial.

OBJECTIVE: The Prevención con Dieta Mediterránea (PREDIMED) trial showed that a Mediterranean diet (MedDiet) supplemented with either extra virgin olive oil or 30 g/d of mixed nuts reduced incident cardiovascular events compared with a control (low fat) diet. The mechanisms of cardiovascular protection afforded by MedDiets remain to be uncovered. We assessed the effect of both supplemented MedDiets on internal carotid intima-media thickness (ICA-IMT) and plaque height, the ultrasound features that best predict future cardiovascular events, in subjects at high cardiovascular risk. **APPROACH AND RESULTS:** In a PREDIMED subcohort (n=175), plaque height and carotid IMT of 3 prespecified segments (ICA, bifurcation, and common) were sonographically assessed at baseline and after intervention for a mean of 2.4 years. We evaluated 164 subjects with complete data. In a multivariate model, mean ICA-IMT progressed in the control diet group (mean [95% confidence interval], 0.052 mm [-0.014 to 0.118 mm]), whereas it regressed in the MedDiet+nuts group (-0.084 mm [-0.158 to -0.010 mm]; P=0.024 versus control). Similar results were observed for maximum ICA-IMT (control, 0.188 mm [0.077 to 0.299 mm]; MedDiet+nuts, -0.030 mm [-0.153 to 0.093 mm]; P=0.034) and maximum plaque height (control, 0.106 mm [0.001 to 0.210 mm]; MedDiet+nuts, -0.091 mm [-0.206 to 0.023 mm]; P=0.047). There were no changes in ICA-IMT or plaque after the MedDiet+extra virgin olive oil. **CONCLUSIONS:** Compared with a control diet, consumption of a MedDiet supplemented with nuts is associated with delayed progression of ICA-IMT and plaque. The results contribute mechanistic evidence for the reduction of cardiovascular events observed in the PREDIMED trial. **CLINICAL TRIAL REGISTRATION URL:** <http://www.controlled-trials.com>. Unique identifier: ISRCTN35739639.